Audio Reverse Engineering — Definition of Done

Oct 18, 2026 · @Antanas

A piece of work is done only when its evidence is committed: a test, a metric, or a recorded decision. Saying it works doesn't count. This doc applies the PRD at five levels: what the user can do, every change, every requirement, every phase gate and the v1 release.

Scores. Every deliverable is tracked as a 0–100 score per language in the Deliverable Scorecard. A deliverable is done when it scores 100 in each of its languages, which means it meets its criteria below with approved evidence.

What the user can do

The app is complete only when the user can do every agreed action below, start to finish, in the studio, with no developer help. Each row is one acceptance walkthrough: someone follows the *Done when* text on a real song, and it either passes or fails.

Every row starts as **Proposed**. To agree on a row, set Status to Agreed, Changed (after editing the row) or Cut. To argue a row, comment on it. Priority and phase are first suggestions.

ID	The user can…	Done when	Priority	Phase	Status
U1	Import a song by dropping a file (WAV, FLAC, MP3) or pasting a URL	The job appears at once and shows progress for each stage; no terminal needed	Must	1	Proposed
U2	Reopen any earlier job from a library	The library lists title, date and status. Opening a job restores stems, notes and recreations without reprocessing	Must	1	Proposed
U3	Split a song into stems and solo or mute each one against the original	Vocals, bass, drums, synth/other and sfx play in sync with the original; any combination can be soloed or muted	Must	1	Proposed
U4	Download any stem	Each stem saves as a 48 kHz WAV that opens in another DAW at the right length and position	Must	1	Proposed

ID	The user can…	Done when	Priority	Phase	Status
U5	See and hear the notes of each melodic stem as a piano roll	Each pitched stem shows a piano roll (drums show a hit grid by drum type); playback follows the original	Must	1	Proposed
U6	Read the transcription as sheet music that follows playback	The score uses its family's notation (see Instrument families) and its cursor stays within one beat of the audio	Must	1	Proposed
U7	Fix a wrong note and hear the fix	Editing a note in the score or piano roll changes the playback and every export	Should	3	Proposed
U8	Export MIDI and MusicXML for any stem	The files open in MuseScore and in another DAW with the same notes and tempo	Must	1	Proposed
U9	See the song's sections and chords on the timeline	Sections (intro, verse, chorus…) and chord names appear as markers; clicking one jumps playback there	Should	2	Proposed
U10	Get matched synth patches for a stem and audition them against the original	At least 3 candidates per stem, each with a match score, playable side by side with the original	Must	3	Proposed
U11	Save a chosen patch as a preset and reuse it	The preset appears in a preset list and loads into any new track in a later session	Should	3	Proposed
U12	Render a synthetic recreation of the whole song and A/B it with the original	One action renders the recreation; one control switches between original and recreation at the same playhead position	Must	2	Proposed
U13	Swap any layer between the original stem and its recreation, and set its level	Every layer has an original/recreated switch and a fader; changes are heard immediately	Must	2	Proposed
U14	Apply a genre remix (jungle, techno, house) to a song	Choosing a genre renders a remix built from the song's own stems and notes, playable and downloadable	Should	2	Proposed
U15	Bounce the mix or all stems to a file	WAV or FLAC export completes without stalling; true peak ⩽ −0.7 dBFS on every file	Must	2	Proposed

ID	The user can…	Done when	Priority	Phase	Status
U16	See how reliable a job's results are	Each job shows note confidence over time and flags low-confidence regions. Gold songs also show F-measure	Must	1	Proposed
U17	Understand and recover from a failed import	A bad or unsupported file gives a plain-language message naming the problem; other jobs and the studio keep working	Must	1	Proposed
U18	Run the whole process from one command or an AI agent	The CLI/MCP run of U1–U12 produces the same job the UI shows, and it opens in the studio	Must	1	Proposed

Walkthrough rules

- Run it on the release candidate, on the workstation, from a fresh start of the studio.
- Use two songs: one from the gold set and one from outside it, each 2–5 minutes long.
- Record pass or fail per row in the release PR, with screenshots or a short screen recording.
- "The studio" means whichever host ADR 0005 picks, SoundCraft or the web studio. The rows don't change either way.

Instrument families

Each instrument family gets its own transcription logic and its own notation. One generic pitch-to-notes path is not done for any family. Every stem is first labelled with a family, either automatically or by the user, and that label picks the transcriber, the notation and the checks.

Family	How it's transcribed	What the user sees	Done when (proposed)
Piano / keys	Polyphonic note model with sustain-pedal detection; notes split into left and right hand	Grand Staff (treble + bass) with chords stacked, a hand-split point and pedal marks	Note F-measure ≥ baseline + 15; ≥ 90% of notes on the right hand
Guitar	Polyphonic notes plus strum and chord detection; each note assigned to a string and fret	Standard staff + TAB (6 strings, tuning selectable), chord symbols with diagrams	Every TAB position is playable (≤ 4-fret span per chord in the chosen tuning); chord names match on gold songs

Family	How it's transcribed	What the user sees	Done when (proposed)
Bass	Monophonic tracker tuned down to 30 Hz, with octave-error correction; note lengths kept (staccato vs held)	Bass clef (8vb where needed) + 4-string TAB	Octave errors < 5% of notes; onsets within ±30 ms
Drums and kick	No pitch. Onsets classified as kick, snare, hi-hat (open/closed), toms, crash or ride; kick found from low-band onsets and checked against the bass	Percussion staff with a fixed drum map; MIDI on channel 10 with General MIDI notes	Kick, snare and hi-hat onset F-measure each ⩾ 0.85 on gold drums; no pitched notes on the drum staff
Synth lead / pad	Mono or poly depending on the stem; notes go to synth matching (FR-6)	Treble or Grand Staff by range, linked to the matched patch	Same note bar as piano; the patch plays the transcribed notes
Vocals	Its own pipeline: see **Voice**	Melody staff with lyrics under notes, one syllable per note	See Voice
SFX / other	Events only, no notes	Labelled markers on an event lane, never a staff	No staff is generated; events can be clicked

- [ ] The family label can be seen and changed per stem, and changing it re-runs only that stem
- [ ] Each family has its own gold-standard clips (⩾ 3 per family, from approved professional musicians) before its bar is frozen
- [ ] MusicXML export keeps family notation (TAB, drum map, Grand Staff) and opens correctly in MuseScore

Voice

Voice is done when the user can trust what the app says is being sung or spoken, word by word, and see it mapped onto the song's structure. Today one engine (Whisper *base*, Lithuanian forced) produces lyrics, and Edge-TTS with pitch shifting re-sings them. The baseline audit found the re-sung voice reads as speech (F0 spread 23.5 Hz against 150.5 Hz in the original). Voice gets its own pipeline, several STT and TTS engines, and its own checks.

Pipeline stages

Stage	What it does	Candidates (chosen by tournament at Gate 0)	Done when (proposed)
1. Isolate	Vocal stem, then lead vs backing split and de-reverb	MSST vocal and karaoke models; Demucs fallback	Lead stem has no audible backing on gold trio songs
2. Detect	Each segment labelled as sung, spoken, rap or no voice; language per line	Silero VAD plus a sung/spoken classifier; Whisper language ID	Segment labels ≥ 90% correct; language per line ≥ 95%
3. Understand (STT)	Several engines transcribe each line, then a consensus with per-word confidence	Cloud STT services, Whisper large-v3 / turbo, faster-whisper, a Lithuanian-tuned model. Later: local-only engines	Word error rate on gold lyrics ⩽ baseline − 30% relative, measured per language
4. Align	Words, then syllables, then phonemes, timed to the audio	WhisperX, MMS forced aligner, Montreal Forced Aligner	Median word-boundary error ⩽ 80 ms against spot-checked timings
5. Map to music	Each syllable tied to a note (melismas kept), notes tied to bars, bars to sections	In-house, using the F0 notes and beat grid	≥ 90% of syllables on the right note in gold songs
6. Identify	Which singer sings each line; how many voices	pyannote diarization plus F0 range per voice	Voice per line ≥ 90% correct on trio songs
7. Re-voice	Speech by TTS, singing by a singing-voice synth, optional voice conversion	Cloud TTS and singing services, Edge-TTS (current), singing-voice synth (DiffSinger class), RVC-class conversion	Re-sung line follows the melody: F0 spread within 20% of the original
8. Evaluate	WER, boundary error, syllable-note accuracy and voice ID per job	`accuracy_evaluator.py`, extended	Metrics stored per job; CI fails on regression

Voice actions to agree on (same rules as What the user can do)

ID	The user can…	Done when	Priority	Status
V1	Read the lyrics with each word highlighted as it is sung	Word highlighting stays within 100 ms of the audio across the whole song	Must	Proposed
V2	See which parts are sung, spoken, rapped or instrumental	A voice lane on the timeline shows the four labels, colour-coded	Must	Proposed
V3	See and correct the language of each line	English and Lithuanian are detected and transcribed; other languages are flagged as unsupported. Changing a line's language re-transcribes only that line	Must	Changed
V4	See which singer sings each line	Lines are tagged Voice 1/2/3 (renamable), and harmonies show who joins	Must	Proposed
V5	Fix a misheard word	Editing a word re-aligns that line in seconds, without reprocessing the song	Must	Proposed
V6	See lyrics under the notes in the score	One syllable per note, melisma lines drawn; Lithuanian syllables and diacritics correct	Must	Changed
V7	Compare what different STT engines heard and pick per line	Engine outputs side by side with confidence; the choice is saved	Should	Proposed
V8	Export lyrics	Plain text, timed LRC (word level) and lyrics inside MusicXML and MIDI	Must	Proposed
V9	Re-sing or re-speak a line with a synthetic voice and A/B it	The new line keeps the original melody and timing and switches with the original at the same playhead	Should	Proposed

ID	The user can…	Done when	Priority	Status
V10	Search the library by what is sung	Typing a word finds songs and jumps to the moment it is sung	Could	Proposed
V11	Edit word-timed lyrics directly on the sheet	On the score the user can retype a word, drag a word or syllable to another note or time, split or merge syllables, and extend or remove a melisma. Highlighting (V1), playback and every export (V8) follow the edit at once. Undo works, and the edit survives reprocessing of other stems	Must	Agreed

Voice decisions (agreed 2026-10-18)

Topic	Decision	Effect on this DoD
Languages	English and Lithuanian as parallel tracks from the start. Other languages later, as language packs	Every voice bar is measured separately for EN and LT, and both must pass for v1
Cloud services	Audio may go to outside STT/TTS services now. A local-only path comes later	Cloud engines can take part in the stage 3 and 7 tournaments. Each job records which service processed it
Legal use	The user takes full responsibility for using the tool legally, including voice cloning and copyrighted audio	No in-app consent or rights checks. The terms of use state this
Gold standards	Only approved professional musicians provide gold-standard sheet music and lyrics	Every gold file records who wrote it and who approved it. Unapproved files don't count toward any bar
Lyric editing	The user must be able to edit word-timed lyrics manually on the sheet	Added as V11 (Must)

Every change

Every commit or PR that touches `apps/remix_studio`, `packages/audiocore` or the host must meet all of these.

- [ ] Scoped commit message (`remix:`, `core:`, `docs:`) and no files outside the change's scope
- [ ] `uv run pytest -m "not slow and not artifact"` green in every app the change touches
- [ ] `tests/test_render_fingerprints.py` green. If a fingerprint changed, the sound change was intended, someone listened to it, and the commit says so
- [ ] New behaviour has at least one hermetic test that fails on bad output (stub models or tiny fixtures, no network)
- [ ] No absolute paths. New locations are added to `studio_paths.py`
- [ ] No ML dependency in `audiocore` or `apps/radio`. Apps don't import each other
- [ ] Bad input (NaN, out-of-range values, corrupt or empty audio) returns a typed error, never a crash or silent output
- [ ] Every rendered file has true peak ≤ −0.7 dBFS
- [ ] Any new dependency or model checkpoint has its licence recorded and is on the allowlist (MIT, BSD, Apache-2.0)
- [ ] Docs updated where behaviour changed: `AGENTS.md`, `README.md`, or an ADR for decisions

Per requirement

Each FR (numbered as in the PRD) is done when its criterion holds and its evidence is committed. Thresholds marked *baseline* are frozen at Gate 0.

ID	Done when	Evidence
FR-1 Ingest	File and URL inputs produce 48 kHz audio plus a manifest with input hash, model names and weight hashes. Re-running the same input reuses the cache	Hermetic test on a 5 s fixture; manifest schema test

ID	Done when	Evidence
FR-2 Separation	MSST writes 5 phase-aligned 48 kHz stems. Their sum stays within −40 dB of the input. Vocal SDR is at least the Demucs 6s baseline. If MSST fails, Demucs runs instead	`slow` test on gold reference stems; SDR table in the catalog
FR-3 Pitch	RMVPE and torchcrepe return F0 and confidence per 10 ms frame. A synthetic sine sweep is tracked within ±20 cents	Hermetic sweep test with stub weights; `slow` test with real weights
FR-4 Transcription	Every melodic stem gets a `.mid` and a `.musicxml` that music21 re-parses into the same notes. The Grand Staff SVG still renders	Round-trip test; gold-standard F-measure ⩾ baseline + 15
FR-5 Structure	Sections and chords are written as markers with start ⩽ end and in-range times. Chord accuracy is measured on gold songs	Hermetic marker-validation test; accuracy row in catalog
FR-6 Synth matching	Returns the top-N patches, every parameter in [0, 1], each with its MR-STFT loss. The best patch beats the hand-modelled timbre on ⩾ 80% of held-out clips	Hermetic test with a stub estimator; held-out evaluation report
FR-7 Session assembly	One call builds tracks, clips, patches and markers. Partial failures come back listed, existing media is never overwritten, and running it twice is idempotent	Agent-task test against the host (or the session-folder fallback)
FR-8 Score view	MusicXML renders and follows the playhead within one beat. Edits round-trip into the MIDI clip	Manual check recorded in the PR with a screenshot; round-trip test

ID	Done when	Evidence
FR-9 Evaluation	Every job on a gold song writes F-measure, SDR and timing to SQLite. CI fails on a drop of more than 2 points	CI job log; catalog rows
FR-10 Interfaces	Every stage runs from CLI and JSON with the same arguments. Errors have a code and a message, never a stack trace	Contract tests per stage; error-case tests
FR-11 Instrument families	Every stem carries a family label that the user can change. Each family meets its bar in Instrument families	Hermetic test per family; per-family metrics in the catalog
FR-12 Voice understanding	Stages 1–6 in Voice meet their bars on English and on Lithuanian gold lyrics, each. Each job records the services used	STT tournament report; recorded cloud responses in hermetic tests; voice metrics in the catalog
FR-13 Lyric editing	V11 passes its walkthrough; edits persist in the job and survive reprocessing of other stems	Round-trip test (edit, save, reload, export); walkthrough record
FR-14 Re-voicing	Stage 7 bar met: a re-sung line's F0 spread is within 20% of the original, with the same timing	Fingerprint test for each local engine; A/B listening note in the PR
FR-15 Gold provenance	Every gold file names its author and approving professional musician; files without approval are skipped by evaluation	Schema test on gold files; evaluation test that ignores unapproved files
FR-16 Gold data sourcing	Worker runs unattended for 7 days; every EN and LT gap has ≥ 3 candidates with V ≥ 60; ≥ 80% of V ≥ 85 candidates are approved over the first 50 reviews; R1–R7, R9 and R10 pass	Worker run log; review outcomes in the catalog; walkthrough records

ID	Done when	Evidence
FR-17 Score and performance	Every job stores score layer, performance layer and tempo map. Median beat error ⩽ 50 ms against spot-checked anchors; ⩾ 90% of written durations recovered by quantisation on gold; expression differences (scorecard table) are not counted as errors; a musician can mark performed variants	Layer schema test; anchor files per gold song; quantisation report
FR-18 Batch alignment	Unit index L1–L10 stored per job; route correct on all 3 LT songs; bar boundaries ⩽ 50 ms median and chord spans ⩾ 85% correct against musician-checked songs; note F computed only on written and embellished units; one song per language timed fully by ear	Unit-index schema test; musician-marked routes and modes; benchmark timing files
FR-19 Loops and mixes	M1–M5 pass their walkthroughs (M6–M8 if agreed); 20 random bar and phrase loops play with no audible click; exports carry tempo, key and chords; true peak ⩽ −0.7 dBFS	Loop listening test record; walkthrough records

Per phase gate

A phase closes only when every box under its gate is ticked. Its work items still have to meet the two sections above.

Gate 0 — Foundation and spike

- [ ] `torch.cuda.is_available()` is True in the remix env, and the CPU path still passes the test suite
- [ ] At least one hermetic test per pipeline stage (ingest, separation, pitch, transcription, evaluation)
- [ ] Scorecard built and scoring 100 itself; its first run stored as the baseline for every deliverable in EN and LT

- [ ] SoundCraft spike report lists each command the PRD relies on as works, missing or broken
- [ ] ADR 0005 records the host decision (SoundCraft, or Python studio plus web UI)
- [ ] Brief PDF committed under `docs/remix/` and the open questions answered or reassigned
- [ ] Approved gold standards: approval records (author, approving musician, scope, checksum) written for Grikiai, Virusas and Prie Topolio; musicians mark each song's form route and improvised spans; one LT song timed fully by ear as the benchmark; first EN gold songs commissioned or sourced

Gate 1 — Sidecar upgrade

- [ ] FR-1, FR-2, FR-3, FR-4, FR-9 and FR-10 meet their criteria
- [ ] Note F-measure ⩾ baseline + 15 on the gold songs
- [ ] The existing studio (`server.py` , remixes) still works, and the render fingerprints are green or deliberately regenerated

Gate 2 — Session assembly

- [ ] FR-5 and FR-7 meet their criteria
- [ ] A 3-minute song becomes an openable session in ⩽ 10 min on the GTX 1660
- [ ] Running the same input twice gives an identical session apart from timestamps

Gate 3 — Synth matching and UI

- [ ] FR-6 and FR-8 meet their criteria
- [ ] Every v1 success metric in the PRD is met (next section), and every Agreed action passes its walkthrough

v1 release

v1 is done when all four gates are closed, every Agreed Must action in What the user can do and in Voice passes its walkthrough, every instrument family meets its bar, and every row below passes on the same commit.

Criterion	Threshold	Measured by
Note F-measure (onset ±50 ms, pitch ±50 cents)	Baseline + 15 points	`gold_standard_benchmark.py`
Vocal stem SDR	⩾ Demucs 6s baseline	Evaluation job on gold reference stems

Criterion	Threshold	Measured by
Synth-match win rate	⩾ 80% of held-out clips	FR-6 evaluation report
Track to session	⩽ 10 min for 3 min of audio	Timed run on the GTX 1660
True peak on every render	⩽ −0.7 dBFS	`audiocore.dsp.true_peak_dbfs` check in tests
Gold-standard coverage	⩾ 8 songs per language (EN and LT), ⩾ 2 genres, each signed off by an approved professional musician	Count in `gold_standards/`
Test tiers	Hermetic, `slow` and `artifact` all green	`uv run pytest` full run log
Licences	Every dependency and checkpoint on the allowlist; no AGPL	Licence table in the repo
Docs	README run steps work on a fresh clone; ADRs current	One fresh-clone dry run

The radio app isn't part of this release. Its VPS deploy stays on its own track under ADR 0002.

How to verify

Run these from `apps/remix_studio` unless noted. Paste the output, or link to it, in the PR.

Check	Command	Evidence lives in
Hermetic tier	`uv run pytest -m "not slow and not artifact"`	PR description
Sound unchanged	`uv run pytest tests/test_render_fingerprints.py`	PR description; `tests/golden/` diff if regenerated
Full suite	`uv run pytest` (needs `data/remix` populated)	Gate checklist
Shared core	`cd packages/audiocore && uv run pytest`	PR description

Check	Command	Evidence lives in
Accuracy	`uv run python gold_standard_benchmark.py`	SQLite catalog `data/remix/db/remix_studio.db`
GPU available	`uv run python -c "import torch; print(torch.cuda.is_available())"`	Gate 0 checklist
Decisions	New ADR in `docs/adr/`	Git history
Scores	uv run python -m scorecard run, then uv run python -m scorecard report	SQLite tables score_runs and scores; the report linked in the PR